

Skills Summary

Multiplication Three Ways

Use this reference while completing your worksheet or when studying.

Skill 1 — From repeated addition to a product

Rule: Count how many times the same number repeats.
that count $\times$ the repeated number = the total.

Example: Write $7 + 7 + 7$ as a multiplication fact and find the total.

Step 1. The 7 repeats 3 times, so the fact is 3×7 .

Step 2. $7 + 7 + 7 = 21$, so $3 \times 7 = 21$.

$\Rightarrow$ **21**

Try it: Write $5 + 5 + 5 + 5$ as a multiplication fact and find the total.

check: $4 \times 5 = 20$

Skill 2 — From an array to a product

Rule: rows $\times$ number in each row = total.
An array is repeated addition you can see: every row adds the same number again.

Example: An array has 4 rows with 9 dots in each row. Write its fact and total.

Step 1. Each of the 4 rows holds 9 dots, so the fact is 4×9 .

Step 2. $4 \times 9 = 36$.

$\Rightarrow$ **36 dots**

Try it: An array has 6 rows with 3 dots in each row. What is its fact and total?

check: $6 \times 3 = 18$

Skill 3 — From a product back to repeated addition

Rule: 3×8 means “3 groups of 8” — write one 8 for each group.
Turning the fact around (8 groups of 3) changes the picture, never the total.

Example: Write 3×8 as repeated addition and find the total.

Step 1. Three groups of 8 means the 8 is added 3 times: $8 + 8 + 8$.

Step 2. $8 + 8 = 16$, and $16 + 8 = 24$.

$\Rightarrow$ **24**

Try it: Write 2×9 as repeated addition and find the total.

check: $6 + 6 = 12$

(!) Watch out: The multiplication sign never means “add the two numbers.” A fact like 6×9 counts 6 whole groups of 9 — the answer is much bigger than $6 + 9$.